CS24351 — ELECTIVE (3.0 CR) | FULL COURSE MASTER | BIT MESRA | NEP SCHEME

Natural Language Processing (CS24351) — Full Course Master Book

Complete End-to-End B.Tech CSE 5th Semester Study Book & PYQ Bank

EXECUTIVE MASTER STUDY GUIDE & LANGUAGE MODELS BANK

Natural Language Processing (CS24311)

Birla Institute of Technology, Mesra | B.Tech CSE 5th Semester (NEP 2024–25 Scheme)

📚 Complete Course Structure & NLP Architecture Matrix

Module	Core Syllabus Scope	Key Algorithms & Formulations
Module I	Morphology & Language Modeling	Regular Expressions, Porter Stemmer, Byte-Pair Encoding (BPE), N-grams, Perplexity, Laplace, Good-Turing & Kneser-Ney Smoothing
Module II	Part-of-Speech Tagging & HMM	Penn Treebank Tagset, Hidden Markov Models (HMM), Forward Algorithm, Viterbi Dynamic Programming Decoding
Module III	Syntactic Parsing & Grammars	Context-Free Grammars (CFG), Chomsky Normal Form (CNF), CYK Parsing Algorithm, Treebanks & Probabilistic CFG (PCFG)
Module IV	Semantics & Word Embeddings	WordNet Synsets, TF-IDF Vectorization, Pointwise Mutual Information (PMI), Word2Vec (Skip-Gram & CBOW), GloVe
Module V	Transformers, LLMs & Generation	Scaled Dot-Product Attention, Multi-Head Attention, Positional Encodings, BERT (Masked LM), GPT, BLEU Score & ROUGE

🎯 Exam Preparation & High-Yield Strategy

This publication-grade master book consolidates all 5 modules with formal mathematical derivations, KaTeX-rendered language model smoothing formulas, Viterbi trellis dynamic programming traces, and model answers to BIT Mesra end-semester examination questions.

Module I: Linguistic Foundations & Preprocessing — Topics Covered

1. Five Levels of Linguistic Analysis
2. Ambiguity in Natural Language (4 Forms)
3. Tokenization & Byte-Pair Encoding (BPE)
4. Porter Stemmer vs. WordNet Lemmatization
5. Regular Expressions for NLP Information Extraction

1. Five Levels of Linguistic Analysis

Linguistic Level	Core Domain of Study	Example Task / Challenge
1. Phonetics / Phonology	Acoustic sounds, phonemes, and pronunciation rules.	Speech-to-Text acoustic modeling.
2. Morphology	Structure of words, morphemes, prefixes, suffixes, and inflections.	Stemming (`running` $\rightarrow$ `run`).
3. Syntax	Grammatical structure, phrase groupings, and word order.	POS Tagging and Parse Trees.
4. Semantics	Literal meaning of words, phrases, and sentence composition.	Word Sense Disambiguation (WSD).
5. Pragmatics / Discourse	Contextual meaning, speaker intent, coreference, and sarcasm.	Anaphora resolution (`Mary saw the car. She bought it.`).

Module II: Language Models, Smoothing & Parsing — Topics Covered

1. N-gram Language Models & Markov Assumption
2. Perplexity Evaluation Metric
3. Smoothing: Laplace, Good-Turing, Kneser-Ney
4. HMM POS Tagging & The Viterbi Algorithm
5. Probabilistic Context-Free Grammars (PCFGs)

1. N-gram Language Models & Perplexity

$$P(w_1, w_2, \ldots, w_N) = \prod_{k=1}^{N} P(w_k \mid w_{k-n+1}^{k-1})$$

$$\text{Perplexity}(W) = P(w_1, \ldots, w_N)^{-1/N} = 2^{-\frac{1}{N}\sum \log_2 P(w_i|w_{i-n+1}^{i-1})}$$

2. Smoothing Techniques Comparison

- **Laplace (Add-1) Smoothing:** $P(w_n \mid w_{n-1}) = \frac{C(w_{n-1}, w_n)+1}{C(w_{n-1})+V}$. (Tends to allocate too much probability mass to unseen items).
- **Kneser-Ney Smoothing:** State-of-the-art N-gram smoothing combining absolute discounting with continuation probabilities based on how versatile a word is as a novel continuation.

Module III: Lexical Semantics & Word Embeddings — Topics Covered

1. WordNet Synsets & Lesk Disambiguation
2. TF-IDF Vector Space & Cosine Similarity
3. Word2Vec: Continuous Bag of Words (CBOW)
4. Word2Vec: Skip-Gram with Negative Sampling
5. GloVe Global Vectors & FastText Subwords

1. Word2Vec Architectures

Architecture	Objective Formulation	Key Strength
Continuous Bag-of-Words (CBOW)	Predicts target center word w_t given sum of surrounding context words $w_{t-c}, \dots, w_{t+c}$.	Faster to train; better representation for frequent words.
Skip-Gram with Negative Sampling (SGNS)	Predicts surrounding context words given target center word w_t: $\max \sum \log \sigma(v'_c \cdot v_w) + \sum \mathbb{E}[\log \sigma(-v'_k \cdot v_w)]$.	Superior performance on rare words and fine-grained analogies.

Module IV: Neural Sequence Models & Transformers — Topics Covered

1. Recurrent Neural Networks (RNN) & BPTT
2. Long Short-Term Memory (LSTM) & GRU Gating
3. Seq2Seq Architecture & Attention Mechanisms
4. The Transformer: Scaled Dot-Product & Multi-Head
5. Pretrained Models: BERT (MLM) vs. GPT (Autoregressive)

1. The Transformer Attention Formulation (Vaswani et al., 2017)

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad \text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

🏛 **BIT Mesra Exam Question (10 Marks)**

Problem: Compare BERT and GPT architectures in terms of attention directionality, pre-training objectives, and downstream fine-tuning.

Solution:

- **BERT:** Bidirectional Transformer Encoder. Pretrained on Masked Language Modeling (MLM 15% tokens) + Next Sentence Prediction (NSP). Ideal for classification, NER, QA extraction.
- **GPT:** Unidirectional Autoregressive Decoder (Causal attention mask). Pretrained on Next-Token Prediction. Ideal for text generation, creative writing, and in-context few-shot prompting.

Module V: NLP Applications & Responsible AI — Topics Covered

1. Text Classification & Sentiment Analysis
2. Named Entity Recognition (BiLSTM-CRF)
3. Machine Translation Paradigms & BLEU Score
4. Task-Oriented Dialogue Systems
5. Fairness, Societal Bias & Ethics in NLP

1. Machine Translation Evaluation: BLEU Metric

BLEU (Bilingual Evaluation Understudy) computes modified n-gram precision penalized by brevity:

$$\text{BLEU} = \text{BP} \times \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \le r \end{cases}$$

where c is candidate translation length and r is reference corpus length.